

Complete Compendium: Cauchy Stress Tensor

From a 5-year-old's intuition, through calculation examples, to differential equations

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1 Introduction and Philosophy of Continuum Mechanics

1.1 Explain Like I'm 5 (ELI5)

Imagine you are building a tower out of LEGO bricks. If you press down on the very top, you feel the brick resisting under your finger. But what happens to the brick right in the middle of the tower? It is also being squished by the ones above it! In mathematics and physics, instead of bricks, we imagine "jelly". We do not talk about individual molecules (because there are too many of them), but we pretend that the jelly is continuous and fills the entire space. We divide forces into two types: those that pull on the entire mass (like Earth's gravity pulling your whole body), and those with which you push or scratch the surface of this jelly.

1.2 Academic Perspective

In mathematical Continuum Mechanics, a body is represented as a set with measure $\Omega \subset \mathbb{R}^3$. Forces are divided into:

- **Body (mass) forces $\mathbf{f}$:** Acting on the entire measure of volume (e.g., gravitational field) $[N/m^3]$.
- **Surface forces $\mathbf{t}$:** Acting on the boundary $\partial\Omega$ or imaginary internal cross-sections $[N/m^2]$.

2 Basic Intuition: Compression vs. Shear

2.1 Explain Like I'm 5

We take a cube of playdough and put it on a table:

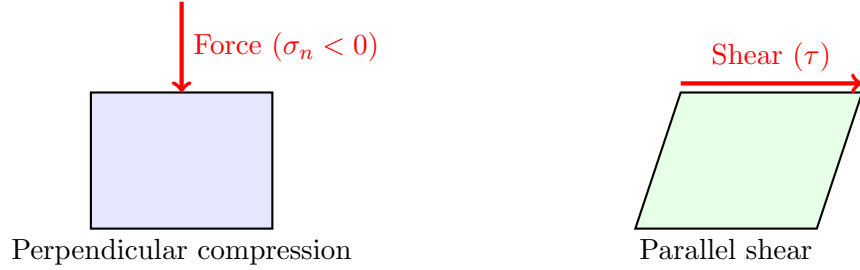
1. **Compression/Tension (Normal):** You place your hand on top of the playdough and press it vertically into the table. The playdough flattens like a pancake. This is an action perpendicular to the surface.
2. **Shear (Cutting):** You place your hand on top of the playdough and slide your hand sideways (along the table). The playdough tilts to the side like the Leaning Tower of Pisa. This is a movement parallel to the surface – exactly the same as spreading butter on bread.

2.2 Academic Perspective and Formalism

For a plane with a normalized normal vector $\mathbf{n}$ ($\|\mathbf{n}\| = 1$), the stress vector $\mathbf{t}^{(\mathbf{n})}$ is decomposed into an orthogonal and a tangential component:

$$\mathbf{t}^{(\mathbf{n})} = \sigma_n \mathbf{n} + \boldsymbol{\tau}$$

- **Normal Stress** ($\sigma_n = \mathbf{t}^{(n)} \cdot \mathbf{n}$): Orthogonal to the plane. $\sigma_n > 0$ means tension, $\sigma_n < 0$ means compression.
- **Shear Stress** ($\boldsymbol{\tau} = \mathbf{t}^{(n)} - \sigma_n \mathbf{n}$): Tangential to the plane, responsible for shape deformation (shearing).



3 Mathematical Formulation: Cauchy's Stress Theorem

3.1 Explain Like I'm 5

Imagine that you are hanging inside the jelly as a miniature little person. Depending on which way you turn and what invisible wall you put in front of you, the force you feel from the surrounding jelly will be different! The Cauchy tensor is a "magic machine" (a 3×3 matrix). You throw a vector indicating the direction of your wall ($\mathbf{n}$) into the machine, and the machine spits out the exact force vector ($\mathbf{t}$) acting on that wall.

3.2 Academic Formula

$$\mathbf{t}^{(n)}(\mathbf{x}) = \boldsymbol{\sigma}(\mathbf{x}) \cdot \mathbf{n}$$

Where $\boldsymbol{\sigma} \in \mathbb{R}^{3 \times 3}$ is the Cauchy stress tensor:

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$

From the conservation of angular momentum follows symmetry: $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$ (meaning $\sigma_{xy} = \sigma_{yx}$, etc.).

4 CAUSES: Where does the tensor $\boldsymbol{\sigma}$ come from?

4.1 Explain Like I'm 5

Why do any forces appear inside the playdough at all?

1. Because someone is squeezing it from the outside with their hands (external loading).
2. Because we stretched it like a rubber band – the molecules want to return to their place, so they pull each other (elastic deformation).
3. Because we heated it up! If you heat a closed can of gas, the molecules will start elbowing the walls (thermal stresses).

4.2 Formalism and Calculation Example of Hooke's Law

In the linear theory of elasticity, the connection between causes (strains $\boldsymbol{\varepsilon}$) and effects (stresses $\boldsymbol{\sigma}$) is described by **Hooke's Law**:

$$\boldsymbol{\sigma} = \lambda \text{tr}(\boldsymbol{\varepsilon}) \mathbf{I} + 2\mu \boldsymbol{\varepsilon}$$

where λ, μ are the Lamé constants characteristic of a given material.

Scenario Calculation (Stretching a rubber cube):

Assume a material with constants $\lambda = 100 \text{ MPa}$, $\mu = 50 \text{ MPa}$. The cube was stretched only along the X axis, which gives the strain tensor:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} 0.02 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Calculation:

$$1. \text{tr}(\boldsymbol{\varepsilon}) = 0.02 + 0 + 0 = 0.02.$$

$$2. \boldsymbol{\sigma} = 100 \cdot 0.02 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + 2 \cdot 50 \cdot \begin{bmatrix} 0.02 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ MPa}.$$

Result Interpretation: Stretching along X caused a stress of 4 MPa in the X axis, but as much as 2 MPa in the Y and Z axes, because the material would like to shrink transversely, but the joints are holding it back!

Scenario 2 Calculation (Complex 3D deformation – full matrix):

We have the same elastic material ($\lambda = 100 \text{ MPa}$, $\mu = 50 \text{ MPa}$). The element was subjected to triaxial tension/compression and shearing in all planes. The strain tensor is:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} 0.01 & 0.005 & -0.002 \\ 0.005 & -0.004 & 0.003 \\ -0.002 & 0.003 & 0.001 \end{bmatrix}$$

Step 1: Calculating the trace of the strain tensor (volumetric strain):

$$\text{tr}(\boldsymbol{\varepsilon}) = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz} = 0.01 + (-0.004) + 0.001 = 0.007$$

Step 2: Applying Hooke's Law ($\boldsymbol{\sigma} = \lambda \text{tr}(\boldsymbol{\varepsilon})\mathbf{I} + 2\mu\boldsymbol{\varepsilon}$): Spherical (hydrostatic) part:

$$\lambda \text{tr}(\boldsymbol{\varepsilon})\mathbf{I} = 100 \cdot 0.007 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.7 & 0 & 0 \\ 0 & 0.7 & 0 \\ 0 & 0 & 0.7 \end{bmatrix} \text{ MPa}$$

Deviatoric (shape deformation) part:

$$2\mu\boldsymbol{\varepsilon} = 2 \cdot 50 \cdot \begin{bmatrix} 0.01 & 0.005 & -0.002 \\ 0.005 & -0.004 & 0.003 \\ -0.002 & 0.003 & 0.001 \end{bmatrix} = \begin{bmatrix} 1.0 & 0.5 & -0.2 \\ 0.5 & -0.4 & 0.3 \\ -0.2 & 0.3 & 0.1 \end{bmatrix} \text{ MPa}$$

Step 3: Summing the matrices ($\boldsymbol{\sigma}$):

$$\boldsymbol{\sigma} = \begin{bmatrix} 0.7 + 1.0 & 0.5 & -0.2 \\ 0.5 & 0.7 - 0.4 & 0.3 \\ -0.2 & 0.3 & 0.7 + 0.1 \end{bmatrix} = \begin{bmatrix} 1.7 & 0.5 & -0.2 \\ 0.5 & 0.3 & 0.3 \\ -0.2 & 0.3 & 0.8 \end{bmatrix} \text{ MPa}$$

Result Interpretation (Step-by-step):

- **Diagonal** ($\sigma_{xx} = 1.7, \sigma_{yy} = 0.3, \sigma_{zz} = 0.8$): All values are positive, which means the material is stretching along every axis. Even though the input strain in the Y axis was negative ($\varepsilon_{yy} = -0.004$), the huge overall volume increase ($\text{tr}(\boldsymbol{\varepsilon}) > 0$) and stretching in X and Z caused such great material resistance that the stress in the Y axis also became tensile!

- **Off-diagonal elements** ($\sigma_{xy} = 0.5, \sigma_{xz} = -0.2, \sigma_{yz} = 0.3$): The off-diagonal elements are exactly the shear values scaled by 2μ . They show how the walls of the imaginary cube are trying to slide (shear) under the influence of complex twisting.

Scenario 3 Calculation (Limit state – von Mises criterion):

We are designing a steel element with a yield strength of $\sigma_y = 250$ MPa. At a certain point in the structure, we know from numerical simulation that the stress tensor depends on an unknown load parameter p (in MPa):

$$\sigma(p) = \begin{bmatrix} 2p & p & 0 \\ p & -p & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Task: Determine the maximum allowable value of parameter $p_{\max}$ at which the steel will not exceed the yield limit (according to the von Mises hypothesis).

Step 1: Formula for von Mises equivalent stress (σ_{vm}): For a plane stress state (when the Z row/column contains zeros):

$$\sigma_{vm} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\sigma_{xy}^2}$$

Step 2: Substituting tensor components dependent on p :

$$\sigma_{xx} = 2p$$

$$\sigma_{yy} = -p$$

$$\sigma_{xy} = p$$

Substituting into the formula:

$$\sigma_{vm} = \sqrt{(2p)^2 - (2p)(-p) + (-p)^2 + 3(p)^2}$$

$$\sigma_{vm} = \sqrt{4p^2 + 2p^2 + p^2 + 3p^2} = \sqrt{10p^2} = p\sqrt{10}$$

Step 3: Equating to the yield strength ($\sigma_{vm} \leq \sigma_y$):

$$p\sqrt{10} \leq 250 \text{ MPa} \implies p_{\max} = \frac{250}{\sqrt{10}} \approx \frac{250}{3.162} \approx 79.06 \text{ MPa}$$

Result interpretation and determining the allowable stress matrix: The maximum load parameter is $p_{\max} \approx 79.06$ MPa. After inserting this value into the stress tensor, we get the maximum stress state this steel will survive:

$$\sigma_{\max} = \begin{bmatrix} 158.12 & 79.06 & 0 \\ 79.06 & -79.06 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ MPa}$$

Engineering Conclusion: The steel withstands a massive 158.12 MPa in tension along X , but only because the shear stress is 79.06 MPa. If there were no shearing ($\sigma_{xy} = 0$), the steel would withstand much more in the X axis! It is the presence of shearing that most quickly leads to the plastic destruction of the metal structure.

5 EFFECTS: Cauchy's Equation of Motion

5.1 Explain Like I'm 5

Think about water in a swimming pool. If the water pressure on your left side is EXACTLY THE SAME as on the right, you will stand still. But if someone creates a wave on the left side (higher stress), the difference in forces will push you to the right! Cauchy's equation says exactly this: If the

stresses on one side of a small block are different than on the other, that block will start to accelerate (fly sideways).

5.2 Differential Formulation and Calculation Scenario

Newton's Second Law for a continuum takes the form:

$$\operatorname{div} \boldsymbol{\sigma} + \mathbf{f} = \rho \mathbf{a}$$

where $\operatorname{div} \boldsymbol{\sigma} = \left[\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z}, \dots \right]^T$.

Calculation Scenario (Will the beam start moving?):

In a certain body with density $\rho = 2 \text{ kg/m}^3$, there are no body forces ($\mathbf{f} = \mathbf{0}$). The stress field in space is:

$$\boldsymbol{\sigma}(x, y, z) = \begin{bmatrix} 3x^2 & 2y & 0 \\ 2y & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Let's check the acceleration $\mathbf{a} = (a_x, a_y, a_z)$ at point $(x, y, z) = (1, 2, 0)$.

Calculations:

$$\operatorname{div} \boldsymbol{\sigma} = \begin{bmatrix} \frac{\partial(3x^2)}{\partial x} + \frac{\partial(2y)}{\partial y} + \frac{\partial(0)}{\partial z} \\ \frac{\partial(2y)}{\partial x} + \frac{\partial(0)}{\partial y} + \frac{\partial(0)}{\partial z} \\ 0 \end{bmatrix} = \begin{bmatrix} 6x + 2 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{At point } (1, 2, 0): \operatorname{div} \boldsymbol{\sigma} = \begin{bmatrix} 6(1) + 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \\ 0 \end{bmatrix}.$$

$$\text{From the equation of motion: } \rho \mathbf{a} = \begin{bmatrix} 8 \\ 0 \\ 0 \end{bmatrix} \implies 2 \cdot \mathbf{a} = \begin{bmatrix} 8 \\ 0 \\ 0 \end{bmatrix} \implies \mathbf{a} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} \text{ m/s}^2.$$

Result Interpretation: The point has an uneven stress distribution in the X axis, which generates a resultant differential force. The body at this point accelerates along the X axis with a value of 4 m/s^2 .

6 Where to get data for the strain tensor?

6.1 Explain Like I'm 5

Imagine you drew a perfect grid of equal squares with a marker on a stretchy rubber glove. When you put the glove on and start stretching your hand in different directions, your squares will change into rectangles, diamonds, or simply tilted rhombuses.

The strain tensor does not need to know what the rubber is made of or with what force you are pulling it (it does not care about Young's modulus). Its sole and exclusive task is a precise mathematical measurement: *by how much the sides of each square stretched and how much the angles between them twisted.* All we need are "before" and "after" photos.

6.2 Academic Perspective

The strain tensor is a **purely geometric** object. It describes the local change in the geometry of a body subjected to deformation, mapping points from the reference (undeformed) configuration $\mathbf{X} = (x, y, z)^T \in \Omega_0$ to the current configuration $\mathbf{x} = (x', y', z')^T \in \Omega_t$.

The displacement vector field $\mathbf{u}(\mathbf{X})$ is defined as:

$$\mathbf{u}(\mathbf{X}) = \mathbf{x} - \mathbf{X}$$

In laboratory and simulation practice, data on the field $\mathbf{u}$ is obtained in two ways:

1. Experimental Method: DIC (Digital Image Correlation)

This is the current standard in biomechanics:

- A random, irregular pattern of fine dots (the so-called *speckle pattern*) is applied to the surface of the artificial skin.
- During the stretching of the sample, a high-speed camera records the movement of the dots.
- Vision software tracks pixel sub-component displacements and returns the displacement vector field $\mathbf{u}(x, y, z) = [u_x, u_y, u_z]^T$ for each point of the skin.

2. Numerical / Theoretical Method (FEM):

In computer simulations, the convergent displacement field $\mathbf{u}$ is obtained as a numerical result from geometric discretization (Finite Element Method).

6.3 2. Numerical Method (FEM): Discretization and Numerical Determination of the Tensor

A. Concept and Explanation (ELI5)

Intuitive translation: In the DIC experiment, we learn about displacements by taking photos of a real sample. In a numerical method (e.g., the Finite Element Method – FEM), we divide our computer-generated patch of artificial skin into a mesh of hundreds of thousands of small triangles or squares (called elements).

Instead of continuous mathematical formulas, the computer only knows the positions at the corners of these elements (called nodes). The task of the numerical algorithm is to "translate" the movements of the element's corners into a continuous strain tensor inside it.

B. Step-by-Step Computational Algorithm

In the numerical method, the process of calculating the strain tensor proceeds in 5 main steps:

1. **Domain Discretization (Ω):** Division of geometry into finite elements (e.g., 4-node 2D quadrilaterals – $Q4$).
2. **Displacement Field Approximation:** Determining the value of $u(x, y)$ inside the element using shape functions $N_i(x, y)$:

$$\mathbf{u}(x, y) = \sum_{i=1}^n N_i(x, y) \mathbf{u}_i$$

3. **Numerical Differentiation (B Matrix):** Building the matrix of shape function derivatives $\mathbf{B}$, linking strains with the nodal displacement vector $\mathbf{q}$:

$$\boldsymbol{\varepsilon} = \mathbf{B}\mathbf{q}$$

4. **Integration at Gauss Points:** Calculating the strain tensor not at the boundaries, but at the so-called numerical integration points (Gauss points) inside the element, which minimizes numerical errors.
5. **Result Extrapolation:** Transferring strain results from Gauss points to the nodes for visualization (strain maps).

C. How does it work and why do we determine the tensor this way?

- **Application of the Jacobian Matrix (J):** In physical finite elements, their shapes are sometimes distorted. The Jacobian matrix allows for recalculating simple differentiation from a local model square ($-1 \leq \xi, \eta \leq 1$) to the actual position in space (x, y) .
- **Discrete Displacement Gradient ($\nabla \mathbf{u}$):** Instead of a continuous analytical function, we calculate the matrix product of shape function derivatives and nodal displacements. This is the numerical realization of differential operators $\frac{\partial u_x}{\partial x}, \frac{\partial u_x}{\partial y}$, etc.

D. Limitations and Flaws of the Numerical Method

Although numerical methods allow for simulating complicated geometries of artificial skin, they have significant limitations:

1. **Discretization Errors (Mesh Density):** A finite element mesh that is too coarse underestimates strain gradients in places of high geometry change (e.g., around a cut or skin suture).
2. **Volumetric Locking Phenomenon:** Artificial skin based on hydrogels or elastomers is an incompressible material ($\nu \approx 0.5$). Classic lower-order finite elements exhibit artificial excessive stiffness under deformation, giving incorrect tensor values.
3. **Sensitivity to Element Quality (Mesh Distortion):** When the skin patch undergoes drastic deformations (e.g., 200% stretch), the finite elements flatten severely, causing the Jacobian determinant $\det(\mathbf{J}) \rightarrow 0$ and the numerical algorithm to throw a division by zero error.
4. **Requirement for Constitutive Relations:** Unlike the vision experiment (DIC), the numerical method needs an accurate material model (e.g., Mooney-Rivlin hyperelasticity parameters) to determine displacements, which are themselves subject to measurement error.

6.4 Accounting for Skin Thickness: 3D Strain Tensor Model

In reality, a patch of artificial skin has a specific thickness h (Z axis). During significant stretching in the (X, Y) plane, a **thinning effect** occurs in the material (strain component $\varepsilon_{zz} < 0$), caused by the incompressibility of elastomers/hydrogels ($\nu \approx 0.5$).

Full 3D Tensor

In three-dimensional space, the strain matrix has a dimension of 3×3 and contains 6 independent components (due to symmetry):

$$\varepsilon_{3D} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{bmatrix}$$

where:

- $\varepsilon_{xx}, \varepsilon_{yy}$ – strains in the plane of the skin (tension/compression),
- $\varepsilon_{zz} = \frac{\partial u_z}{\partial z}$ – **change in the thickness of the skin patch**,
- $\varepsilon_{xz}, \varepsilon_{yz}$ – shear strains across the thickness (e.g., when pressing a tactile sensor into the skin).

When is 2D enough, and when MUST you use 3D?

1. **2D Simplification (Plane Strain / Plane Stress State):** Used when we are only studying the stretching of a flat strip of skin in a tensile testing machine, and we determine thickness changes simply from Poisson's analytical formula: $\varepsilon_{zz} = -\frac{\nu}{1-\nu}(\varepsilon_{xx} + \varepsilon_{yy})$.
2. **Full 3D Model:** Necessary when:
 - The skin is multilayered (e.g., epidermis + silicone substrate + flexible copper track).
 - We study the indentation of an element (e.g., needle, robot, or sensor) into the thickness of the skin (the so-called *indentation test*).
 - There is a risk of delamination (peeling of layers) due to shear stresses σ_{xz}, σ_{yz} .

7 How to calculate the Strain Tensor step-by-step?

Depending on how hard the artificial skin is stretched, mathematics offers two formulations:

7.1 Scenario A: Small deformations (Linear theory, e.g., stretching up to 5%)

ELI5: If you only pull the skin gently, the curvature of the squares is so small that we use simple derivatives.

We use the **small Cauchy-Green strain tensor** (ε), which is the symmetric part of the displacement field gradient $\nabla \mathbf{u}$:

$$\varepsilon = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$$

In 3×3 components, it looks like this:

$$\varepsilon = \begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{1}{2} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y} \right) & \frac{\partial u_y}{\partial y} & \frac{1}{2} \left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z} \right) & \frac{1}{2} \left(\frac{\partial u_z}{\partial y} + \frac{\partial u_y}{\partial z} \right) & \frac{\partial u_z}{\partial z} \end{bmatrix}$$

7.2 Scenario B: Large deformations (Hyperelasticity – stretching by 20%, 50% and more)

ELI5: Artificial skin resembles silicone or rubber. If you stretch it hard, simple math starts to lie (it would yield negative volumes or incorrect results). We have to add a quadratic term that accounts for large rotations and shape changes.

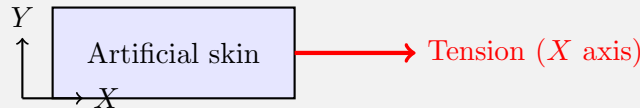
Skin (both natural and artificial) is a **hyperelastic** material. In this case, the **Green-Lagrange Strain Tensor** ($\mathbf{E}$) is used:

$$\mathbf{E} = \frac{1}{2} (\mathbf{F}^T \mathbf{F} - \mathbf{I})$$

where $\mathbf{F} = \nabla \mathbf{x} = \mathbf{I} + \nabla \mathbf{u}$ is the **deformation gradient**.

8 Concrete Calculation Example (Stretching a skin sample)

Imagine a patch of artificial skin subjected to uniaxial tension along the X axis:



From DIC measurements, we know that point (x, y, z) moved according to the field:

- $u_x(x, y, z) = 0.10 \cdot x$ (10% stretch along X)
- $u_y(x, y, z) = -0.03 \cdot y$ (transverse contraction / Poisson effect in Y)
- $u_z(x, y, z) = -0.03 \cdot z$ (thinning of the patch in Z)

Step 1: Calculating partial derivatives:

$$\frac{\partial u_x}{\partial x} = 0.10, \quad \frac{\partial u_y}{\partial y} = -0.03, \quad \frac{\partial u_z}{\partial z} = -0.03$$

All mixed derivatives (e.g., $\frac{\partial u_x}{\partial y}$) are equal to 0.

Step 2: Resulting Strain Tensor ε :

$$\varepsilon = \begin{bmatrix} 0.10 & 0 & 0 \\ 0 & -0.03 & 0 \\ 0 & 0 & -0.03 \end{bmatrix}$$

Step 3: How to get stresses σ from this?

It is only here that the physical properties of the material come into play. If we adopt a linear elasticity model (for small strains) with a skin Young's modulus of $E = 1 \text{ MPa}$ and a Poisson's ratio of $\nu = 0.3$:

$$\sigma = \begin{bmatrix} 1.07 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ MPa}$$

Result Interpretation: A tensile stress of 1.07 MPa was generated along the X axis. In the Y and Z axes, the stresses are 0 MPa because the edges of the material were free and could freely contract ($\varepsilon_{yy} = \varepsilon_{zz} = -0.03$).

8.1 Constitutive Relations: Engineering Perspective vs. Tensor Hooke's Law

To transition from geometry (ε) to forces (σ) in the linear-elastic range, we apply Hooke's Law in two formulations:

A. Engineering Perspective (Intuitive – ELI5)

For uniaxial stretching along the X axis with free edges (e.g., stretching a strip of skin in a machine):

- Principal longitudinal strain: $\varepsilon_{xx} = \frac{\Delta L}{L_0}$
- Transverse strains (Y contraction and Z thinning caused by the Poisson Effect):

$$\varepsilon_{yy} = \varepsilon_{zz} = -\nu \cdot \varepsilon_{xx}$$

- Corresponding longitudinal stress: $\sigma_{xx} = E \cdot \varepsilon_{xx}$ (where E is Young's modulus).

B. 3D Tensor Perspective (Full Continuum Mechanics Formalism)

In the three-dimensional formulation for an isotropic material, Hooke's Law is written using Lamé constants (λ and μ):

$$\sigma = \lambda \text{tr}(\varepsilon)\mathbf{I} + 2\mu\varepsilon$$

Converting from engineering parameters (E, ν) to mathematical Lamé constants (λ, μ):

$$\mu = G = \frac{E}{2(1 + \nu)} \quad (\text{Shear modulus / modulus of rigidity})$$

$$\lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)} \quad (\text{First Lamé constant - handles volume change})$$

9 When will stresses be maximum and the material tear?**9.1 A. Failure Criteria (When will rupture occur?)**

ELI5: Every material has its "yield limit". It is as if the polymer chains inside were small strings – at some point, you will stretch them so much that they will start snapping one after another.

1. Maximum Principal Stress Criterion ($\sigma_1 \geq \sigma_{ult}$):

- Used when the material behaves in a relatively brittle manner or tears through the initiation of a single crack perpendicular to the direction of tension.
- You look for the largest eigenvalue of the Cauchy tensor (σ_1). When σ_1 exceeds the tensile strength σ_{ult} (for human skin it is approx. 5 – 30 MPa depending on the body area), the material tears.

2. Maximum Principal Strain Criterion ($\varepsilon_1 \geq \varepsilon_{max}$):

- For elastomers and hydrogels, it is often not stress, but the **limit strain** that determines the moment of rupture (polymer chains straighten and tear after exceeding, for example, $\varepsilon_1 = 150\%$).

3. Strain Energy Density ($W \geq W_{crit}$):

- For advanced hyperelastic models (Ogden or Mooney-Rivlin models used in artificial skin), the material fractures when the energy stored in the volume exceeds a critical value.

9.2 B. Where and under what conditions do maximum stresses arise?

Maximum stresses (and consequently, rupture) will appear under the following physical conditions:

- **Stress Concentration (Notch / Crack):**

If a skin patch has a cut, seam, or hole, stresses in the vicinity of this point increase locally even 3 to 5 times (the so-called notch effect). In the mathematical model, the divergence and strain gradient reach a local extremum there.

- **Stiffness Mismatch (Substrate vs. Skin):**

If you glue artificial skin to a rigid foundation (e.g., a copper sensor or prosthesis), giant shear stresses (σ_{xz}, σ_{yz}) will arise at the phase boundary during stretching, leading to delamination or tearing.

- **Strain Rate ($\dot{\varepsilon}$):**

Artificial skin is a **viscoelastic** material. The faster you stretch it (large $\frac{d\varepsilon}{dt}$), the stiffer the material behaves and the easier it is to destroy at lower strains.

10 Tensor Decomposition: Hydrostatic and Deviatoric Parts

10.1 Explain Like I'm 5

Imagine a playdough ball.

1. **Hydrostatic Part (Compression):** You put the ball into a deep pool. The water compresses it perfectly equally from all sides. The ball decreases in size, but it still remains a perfect little ball (does not change shape).
2. **Deviatoric Part (Shear):** You crush the ball flat with your hand. The volume of the playdough is exactly the same, but you change its shape. It is the deviator that is responsible for destroying and yielding objects!

10.2 Formulas and Decomposition Example

$$\boldsymbol{\sigma} = \mathbf{s} + p\mathbf{I}$$

where $p = \frac{1}{3}\text{tr}(\boldsymbol{\sigma}) = \frac{\sigma_{xx} + \sigma_{yy} + \sigma_{zz}}{3}$, and $\mathbf{s} = \boldsymbol{\sigma} - p\mathbf{I}$ (deviator, for which $\text{tr}(\mathbf{s}) = 0$).

Step-by-step calculation:

Given the stress tensor:

$$\boldsymbol{\sigma} = \begin{bmatrix} 12 & 4 & 0 \\ 4 & 6 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

1. We calculate the mean (hydrostatic) pressure:

$$p = \frac{12 + 6 + 3}{3} = \frac{21}{3} = 7$$

2. We determine the hydrostatic tensor $p\mathbf{I}$:

$$p\mathbf{I} = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

3. We determine the deviator $\mathbf{s} = \boldsymbol{\sigma} - p\mathbf{I}$:

$$\mathbf{s} = \begin{bmatrix} 12-7 & 4-0 & 0 \\ 4-0 & 6-7 & 0 \\ 0 & 0 & 3-7 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 0 \\ 4 & -1 & 0 \\ 0 & 0 & -4 \end{bmatrix}$$

Correctness check: $\text{tr}(\mathbf{s}) = 5 + (-1) + (-4) = 0$. The result perfectly matches the theory!

11 Spectral Properties (Eigenvalues and Eigenvectors)

11.1 Explain Like I'm 5

When you pull a square handkerchief at a strange angle, the handkerchief wrinkles and shears. But if you rotate the handkerchief and pull it exactly along its diagonal, the handkerchief will simply lengthen without any twisting! Eigenvectors are exactly those "magic directions" along which the material is only stretched or compressed, and not twisted/sheared at all.

11.2 Academic Formalism

From the spectral theorem for symmetric matrices, there exist eigenvalues $\sigma_1 \geq \sigma_2 \geq \sigma_3$ and a basis of orthonormal eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, in which the tensor takes a diagonal form:

$$\boldsymbol{\sigma}' = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$

12 Coordinate System Transformation ($Q\boldsymbol{\sigma}Q^T$)

12.1 Explain Like I'm 5

An object in reality does not know how you drew the X, Y, Z axes in your notebook. The forces in it really exist. If you tilt your head by 45° , what was previously "vertical pressing" for you suddenly becomes diagonal "shearing". Matrix transformation is simply recalculating these same physical forces after rotating your coordinate system.

12.2 Formalism and Rotation Scenario Calculation (45° rotation)

When the coordinate system is rotated, represented by the orthogonal matrix $Q \in SO(3)$:

$$\boldsymbol{\sigma}' = Q\boldsymbol{\sigma}Q^T$$

2D Rotation Scenario Calculation:

We have a state of pure tension along the X axis: $\boldsymbol{\sigma} = \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix}$.

We rotate the coordinate system by an angle $\theta = 45^\circ$. Rotation matrix:

$$Q = \begin{bmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

We calculate $\sigma' = Q\sigma Q^T$:

$$Q\sigma = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 10/\sqrt{2} & 0 \\ -10/\sqrt{2} & 0 \end{bmatrix}$$

$$\sigma' = \begin{bmatrix} 10/\sqrt{2} & 0 \\ -10/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 5 & -5 \\ -5 & 5 \end{bmatrix}$$

Result Interpretation: Attempting to rotate the plane by 45° , pure tension $\sigma_{xx} = 10$ turned into mixed states: 5 MPa tension and a massive shear $\sigma'_{xy} = -5$ MPa!

13 Stress Tensor Invariants

13.1 Explain Like I'm 5

Since the elements of the matrix change when you tilt your head, is there anything that remains UNCHANGED? Yes! Invariants are special numbers calculated from a matrix that are exactly the same no matter how you rotate the coordinate system.

13.2 Definitions and Invariant Calculation

1. $I_1 = \text{tr}(\sigma) = \sigma_{xx} + \sigma_{yy} + \sigma_{zz}$
2. $I_2 = \sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx} - \sigma_{xy}^2 - \sigma_{yz}^2 - \sigma_{zx}^2$
3. $I_3 = \det(\sigma)$

Numerical proof of I_1 invariance for the previous rotation example:

For the initial matrix $\sigma = \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix}$:

$$I_1 = 10 + 0 = 10$$

For the matrix after the 45° rotation, i.e., $\sigma' = \begin{bmatrix} 5 & -5 \\ -5 & 5 \end{bmatrix}$:

$$I'_1 = 5 + 5 = 10$$

Conclusion: $I_1 = I'_1$. Despite a complete change in the values of individual matrix cells, the sum of the trace is exactly the same!

14 Detailed Calculation Example (Summary)

We are given the stress tensor at a point:

$$\sigma = \begin{bmatrix} 10 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ MPa}$$

14.1 Step 1: Stress Vector

For a plane with normal $\mathbf{n} = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right]^T$:

$$\mathbf{t}^{(\mathbf{n})} = \sigma \mathbf{n} = \begin{bmatrix} 10 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 12/\sqrt{2} \\ 7/\sqrt{2} \\ 0 \end{bmatrix}$$

14.2 Step 2: Decomposition into Normal Stress (σ_n) and Shear (τ)

Normal stress (compression/tension):

$$\sigma_n = \mathbf{t}^{(\mathbf{n})} \cdot \mathbf{n} = \left(\frac{12}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) + \left(\frac{7}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) = 6 + 3.5 = 9.5 \text{ MPa}$$

Shear stress (τ):

$$\begin{aligned} \|\mathbf{t}^{(\mathbf{n})}\|^2 &= \left(\frac{12}{\sqrt{2}}\right)^2 + \left(\frac{7}{\sqrt{2}}\right)^2 = 72 + 24.5 = 96.5 \\ \tau &= \sqrt{\|\mathbf{t}^{(\mathbf{n})}\|^2 - \sigma_n^2} = \sqrt{96.5 - (9.5)^2} = \sqrt{96.5 - 90.25} = \sqrt{6.25} = 2.5 \text{ MPa} \end{aligned}$$

15 Questions for Office Hours (Academic Discourse)

This section gathers advanced theoretical questions intended for discussion with the instructor. Each of them touches upon the boundaries of classical continuum mechanics theory and the transition to more advanced formulations (non-classical media, large deformations, PDE boundary conditions).

1. On the Symmetry of the Cauchy Tensor and Polar Media

Explanation and context: The classical Cauchy principle of angular momentum conservation implies the symmetry of the stress tensor ($\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$), assuming the absence of external point moments and the absence of internal microrotations. In materials with a microstructure (e.g., composites, granular media, some biomaterials, and artificial skin with a microfiber network), standard Cauchy mechanics may, however, be insufficient.

Intuition (ELI5): In a classical material, a cut-out cube does not rotate by itself when sheared — the shear forces on opposite walls balance out ($\sigma_{xy} = \sigma_{yx}$). However, when microfibers are embedded inside artificial skin, during deformation they begin to locally **rotate like microscopic screws**. This rotation induces an internal torque, which causes some energy to be accumulated in microrotation, and the classical symmetry of the stress tensor is broken ($\boldsymbol{\sigma} \neq \boldsymbol{\sigma}^T$).

Mathematical-Physical Formalism & Numerical Tools

In the micropolar Cosserat theory, additional microrotational degrees of freedom $\boldsymbol{\theta} \in \mathbb{R}^3$ are assigned to each point. The angular momentum conservation equation then takes into account the couple stress tensor $\mathbf{m}$ and the internal body couple density $\mathbf{c}$:

$$\text{div } \boldsymbol{\sigma} + \mathbf{f} = \rho \mathbf{a} \quad (1)$$

$$\text{div } \mathbf{m} + \boldsymbol{\epsilon} : \boldsymbol{\sigma} + \mathbf{c} = \rho \mathbf{j} \cdot \ddot{\boldsymbol{\theta}} \quad (2)$$

where $\boldsymbol{\epsilon} : \boldsymbol{\sigma}$ represents the contraction with the Levi-Civita tensor, which captures the asymmetry of the Cauchy tensor $\text{vec}(\boldsymbol{\sigma} - \boldsymbol{\sigma}^T)$.

- **FEM Tools:** Implementation requires *Cosserat continuum elements* with 6 degrees of freedom per node (3 translations + 3 rotations) and the introduction of a material characteristic length l_c .

2. On Lagrangian vs. Eulerian Formulations for Large Deformations

Explanation and context: The Cauchy tensor $\boldsymbol{\sigma}$ is defined in the current configuration Ω_t (Eulerian description), relating the current force to the current surface area. With large deformations of artificial skin (e.g., 50 – 100% stretching), tracking the changing geometry Ω_t is computationally difficult, so it is more convenient to relate the stresses to the undeformed reference configuration Ω_0 (Lagrangian description).

Intuition (ELI5): When you stretch a rubber sheet of skin from 10 cm to 20 cm, its thickness drastically decreases. The Eulerian description calculates the stress by dividing the force by what it sees **now** (the reduced cross-section at instantaneous time t). The Lagrangian description relates the same force to the original cross-section from the store shelf ($t = 0$). This allows the numerical mesh to be anchored on a constant, undeformed initial geometry.

Mathematical-Physical Formalism & Numerical Tools

The relations between the current configuration Ω_t and the reference configuration Ω_0 are defined by the deformation gradient tensor $\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}}$ and the volume change $J = \det(\mathbf{F})$. The transition between stress tensors is realized through *pull-back* and *push-forward* operations:

1. **First Piola-Kirchhoff Tensor ($\mathbf{P}$):** Connects the force in the current configuration with the reference surface:

$$\mathbf{P} = J\boldsymbol{\sigma}\mathbf{F}^{-T} \quad (3)$$

2. **Second Piola-Kirchhoff Tensor ($\mathbf{S}$):** A symmetric tensor in the full Lagrangian description (Ω_0):

$$\mathbf{S} = \mathbf{F}^{-1}\mathbf{P} = J\mathbf{F}^{-1}\boldsymbol{\sigma}\mathbf{F}^{-T} \quad (4)$$

- **FEM Tools:** The *Total Lagrangian* (TL) formulation is used, in which all volume integrals and stiffness matrices are calculated over the constant domain Ω_0 using the tensor $\mathbf{S}$ and Green-Lagrange strains $\mathbf{E}$.

3. On the Formalism of PDE Boundary Conditions (Neumann vs. Dirichlet)

Explanation and context: Cauchy's equation of motion $\text{div } \boldsymbol{\sigma} + \mathbf{f} = \rho \mathbf{a}$ constitutes a system of partial differential equations (PDE). To obtain a unique solution (in FEM simulation or analytically), displacement boundary conditions (Dirichlet) and force/stress boundary conditions (Neumann) must be defined on the boundary of the domain $\partial\Omega = \Gamma_D \cup \Gamma_N$ ($\Gamma_D \cap \Gamma_N = \emptyset$).

Intuition (ELI5):

- **Dirichlet (Kinematics):** You rigidly screw the edge of the skin to the table and tell the computer: "*This edge has a displacement $u = 0$ mm*". You impose the position.
- **Neumann (Statics):** You attach a weight to the edge or pull it with a force of 10 N. You tell the computer: "*Such a force acts here*", and it calculates by how many millimeters the edge will move.

Mathematical-Physical Formalism & Numerical Tools

The system of differential equations in strong form takes the form:

$$\begin{cases} \text{div } \boldsymbol{\sigma}(\mathbf{u}) + \mathbf{f} = \rho \ddot{\mathbf{u}} & \text{in } \Omega \\ \mathbf{u} = \bar{\mathbf{u}} & \text{on } \Gamma_D \quad (\text{Dirichlet}) \\ \boldsymbol{\sigma} \cdot \mathbf{n} = \bar{\mathbf{t}} & \text{on } \Gamma_N \quad (\text{Neumann}) \end{cases} \quad (5)$$

Transition to the weak formulation (principle of virtual work) by multiplying by a test function $\mathbf{v}$:

$$\int_{\Omega} \boldsymbol{\sigma} : \nabla \mathbf{v} \, dV = \int_{\Omega} \mathbf{f} \cdot \mathbf{v} \, dV + \int_{\Gamma_N} \bar{\mathbf{t}} \cdot \mathbf{v} \, dS \quad (6)$$

- **FEM Implementation:**

- **Dirichlet Conditions (Γ_D):** These are *essential conditions*. They are applied directly at the level of the stiffness matrix $\mathbf{K}\mathbf{u} = \mathbf{F}$ by modifying/eliminating appropriate degrees of freedom in the system of equations (or by the Lagrange multiplier method / penalty function).
- **Neumann Conditions (Γ_N):** These are *natural conditions*. They automatically appear on the right side of the system as the external load vector $\mathbf{F}_{\text{ext}} = \int_{\Gamma_N} \mathbf{N}^T \bar{\mathbf{t}} \, dS$.

Summary and Cheat Sheet

1. Summary of Forces and Stresses

Concept	Symbol / Formula	Physical / Geometric Interpretation
Surface forces	$\mathbf{t}$ $[N/m^2]$	Acting on the body boundary or internal cross-sections.
Body forces	$\mathbf{f}$ $[N/m^3]$	Acting on the entire mass (e.g., gravity).
Stress vector	$\mathbf{t}^{(\mathbf{n})} = \boldsymbol{\sigma} \mathbf{n}$	Force vector on a plane with normal $\mathbf{n}$.
Normal Stress	$\sigma_n = \mathbf{t}^{(\mathbf{n})} \cdot \mathbf{n}$	Perpendicular to the plane (tension > 0 , compression < 0).
Shear Stress	$\boldsymbol{\tau} = \mathbf{t}^{(\mathbf{n})} - \sigma_n \mathbf{n}$	Tangential to the plane (sliding layers / shearing).

2. Key Tensors and Their Decompositions

- **Cauchy Tensor ($\boldsymbol{\sigma}$):** A symmetric 3×3 matrix ($\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$) describing the stress state at a point.
- **Stress Tensor Decomposition:**

$$\boldsymbol{\sigma} = \mathbf{s} + p\mathbf{I}$$

- **Hydrostatic Part ($p\mathbf{I}$):** $p = \frac{1}{3}\text{tr}(\boldsymbol{\sigma})$ – volume change without shape change.
- **Deviatoric Part ($\mathbf{s}$):** $\text{tr}(\mathbf{s}) = 0$ – shape change (shear), main cause of yielding.

- **Small Strain Tensor ($\boldsymbol{\varepsilon}$):**

$$\boldsymbol{\varepsilon} = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) \quad (\text{for small deformations } < 5\%)$$

- **Green-Lagrange Strain Tensor ($\mathbf{E}$):**

$$\mathbf{E} = \frac{1}{2} (\mathbf{F}^T \mathbf{F} - \mathbf{I}), \quad \mathbf{F} = \mathbf{I} + \nabla \mathbf{u} \quad (\text{for large deformations / skins / elastomers})$$

3. Differential Equations and Constitutive Relations

Law / Equation	Formula	Role in the Model
Cauchy's Equation of Motion	$\text{div } \boldsymbol{\sigma} + \mathbf{f} = \rho \mathbf{a}$	Newton's II law in continuum form (motion/acceleration).
Hooke's Law (Linear 3D)	$\boldsymbol{\sigma} = \lambda \text{tr}(\boldsymbol{\varepsilon}) \mathbf{I} + 2\mu \boldsymbol{\varepsilon}$	Connecting geometry ($\boldsymbol{\varepsilon}$) with forces ($\boldsymbol{\sigma}$).
Coordinate Transformation	$\boldsymbol{\sigma}' = \mathbf{Q} \boldsymbol{\sigma} \mathbf{Q}^T$	Matrix rotation about a system rotated by $\mathbf{Q} \in SO(3)$.

4. Invariants and Failure (Rupture) Criteria

Tensor Invariants (Independent of system rotation):

$$I_1 = \text{tr}(\boldsymbol{\sigma}) = \sigma_1 + \sigma_2 + \sigma_3$$

$$I_2 = \frac{1}{2} ((\text{tr}\boldsymbol{\sigma})^2 - \text{tr}(\boldsymbol{\sigma}^2))$$

$$I_3 = \det(\boldsymbol{\sigma})$$

Material Failure / Yield Criteria:

1. Von Mises (Metals / Shear):

$$\sigma_{vm} = \sqrt{\frac{1}{2} [(\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 6(\sigma_{xy}^2 + \sigma_{yz}^2 + \sigma_{zx}^2)]} \geq \sigma_y$$

2. Maximum Principal Stress (Brittle fracture / Skin):

$$\sigma_1 = \max(\text{eig}(\boldsymbol{\sigma})) \geq \sigma_{ult}$$

3. Maximum Principal Strain (Elastomers / Hydrogels):

$$\varepsilon_1 = \max(\text{eig}(\boldsymbol{\varepsilon})) \geq \varepsilon_{max} \quad (\text{e.g. } 150\%)$$

16 Practical Application: Numerical Modeling of Artificial Skin

In this section, we combine all the accumulated theory – from strain kinematics, through the Cauchy tensor, to differential equations – into a coherent engineering apparatus used for the numerical design and analysis of artificial skin (both in sensory coatings and biomechanical prostheses).

16.1 1. Mapping Theoretical Concepts to Simulation Algorithms (FEM/DIC)

In a computer simulation or camera measurement analysis (DIC), each of the concepts discussed earlier plays a strictly defined role in the computational loop:

1. **Displacement field $\mathbf{u}(x, y, z)$ – Experimental input / Nodal variable:** Both the DIC apparatus and the FEM solver start from a discrete displacement vector of points $\mathbf{u} = [u_x, u_y, u_z]^T$. This is pure geometry without material physics.
2. **Strain tensor ($\boldsymbol{\varepsilon}$ vs $\mathbf{E}$) – Deformation kinematics:**
 - For micro-modeling (e.g., small deflections of a tactile sensor $< 5\%$), we use the small Cauchy strain tensor $\boldsymbol{\varepsilon}$.
 - For large stretchings of the skin patch ($> 20\%$), we use the Green-Lagrange tensor $\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$, avoiding errors of artificial volume change during element rotations.
3. **Constitutive relations ($\boldsymbol{\sigma} = f(\boldsymbol{\varepsilon})$ or $W(C)$) – Material physics:** The algorithm transforms pure strain geometry into internal forces. In the elastic range, it uses Lamé constants (λ, μ) , while in the range of large deformations, it uses the strain energy density (e.g., Mooney-Rivlin or Ogden model).
4. **Cauchy Tensor ($\boldsymbol{\sigma}$) – Internal stress state:** Used to evaluate how the material "suffers" under load. The matrix $\boldsymbol{\sigma}$ is determined at numerical integration points (Gauss points) of each finite element.
5. **Hyperelastic Models: Mooney-Rivlin and Ogden**

In the case of artificial skin subjected to large deformations ($> 20\%$), linear Hooke's Law ceases to apply due to the highly nonlinear characteristics of elastomeric and biological materials. A **strain energy density function** W is then used:

$$\boldsymbol{\sigma} \approx \frac{\partial W}{\partial \mathbf{F}}$$

A. Mooney-Rivlin Model (Moderate deformations up to 100%)

The model is based on the strain tensor invariants (I_1, I_2). In the two-parameter form, the energy density is:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$$

where C_{10}, C_{01} are material constants determined in a tensile test. This is the standard model for silicones (e.g., PDMS, Ecoflex).

B. Ogden Model (Large deformations and biological tissues)

The model is based directly on the principal stretches $\lambda_1, \lambda_2, \lambda_3$ (L/L_0):

$$W = \sum_{i=1}^N \frac{\mu_i}{\alpha_i} (\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3)$$

Thanks to the powers α_i , the model perfectly replicates the stiffening effect of living skin and hydrogels at extreme deformations.

6. **Cauchy's Equation ($\text{div } \boldsymbol{\sigma} + \mathbf{f} = \rho \mathbf{a}$) – Equilibrium Criterion and Dynamics:** It checks whether the balance of forces is maintained at a given point of the skin. If $\text{div } \boldsymbol{\sigma} \neq \mathbf{0}$, a wave acceleration occurs in the object (e.g., response to a sudden impact/damage).

16.2 2. Material Database for Artificial and Natural Skins

To perform a correct numerical simulation, we must supply the model with appropriate physical constants. The table below presents a summary of material parameters commonly used in biomechanics and flexible electronics:

Material / Tissue	Young's Modulus (E)	Poisson's Ratio (ν)	Strength (σ_{ult})	Limit (ε_{max})	Strain
Human skin (Epidermis)	0.1 – 2.0 MPa	0.45 – 0.49	5 – 20 MPa	30% – 70%	
Ecoflex 00-30 Silicone (Artificial skin)	0.06 – 0.12 MPa	≈ 0.499	1.4 MPa	900%	
PDMS (Polydimethylsiloxane)	1.0 – 3.0 MPa	0.48	6 – 10 MPa	40% – 120%	
Polyacrylamide hydrogel	0.01 – 0.05 MPa	≈ 0.50	0.1 – 0.5 MPa	100% – 300%	

Table 1: Summary of mechanical properties of materials used in skin modeling.

Key numerical observation: All of the above materials are characterized by a Poisson's ratio $\nu \rightarrow 0.5$. In FEM simulations, this means an **almost incompressible** material, which requires the use of elements with a specially formulated pressure volume term (mixed u/p formulation) to avoid numerical locking (*volumetric locking*).

16.3 3. Failure Prediction: When will the Skin Withstand, and When will it Tear?

In numerical simulation, material rupture is modeled by **element deletion**, when the limit condition is met.

A. Scenario A: Skin Withstands (Safe State)

The skin remains in an undamaged state when the threshold conditions are met at each Gauss point:

$$\sigma_1(\boldsymbol{\sigma}) < \sigma_{ult} \quad \text{AND} \quad \varepsilon_1(\boldsymbol{\varepsilon}) < \varepsilon_{max}$$

where σ_1, ε_1 are the maximum eigenvalues of the respective tensors.

B. Scenario B: Skin Will Rupture (Crack Initiation)

Rupture most commonly occurs according to two mechanisms:

1. **Exceeding Chain Stretch (ε_1 Criterion):** Elastomers and hydrogels tear when polymer chains straighten completely. If the local $\varepsilon_1 \geq \varepsilon_{max}$, microcracking occurs.
2. **Stress Concentration Around a Notch (σ_{vm}/σ_1 Criterion):** If there is a cut with radius r in the patch, the stress at the tip of the cut is:

$$\sigma_{max} = \sigma_0 \left(1 + 2\sqrt{\frac{a}{r}} \right)$$

where a is the length of the cut, and σ_0 is the nominal tensile stress. When $\sigma_{max} \geq \sigma_{ult}$, the crack propagates uncontrollably through the skin patch.

16.4 4. Comprehensive Calculation Example: Designing a Stretchable Track in Artificial Skin

Problem Description:

We are designing a patch of artificial skin made of PDMS silicone ($E = 2 \text{ MPa}, \nu = 0.48, \sigma_{ult} = 8 \text{ MPa}$), inside which a conductive copper track is embedded ($E_{Cu} = 110 \text{ GPa}$). The skin patch has a thickness of $h = 2 \text{ mm}$ and is stretched along the X axis by 30%.

Step 1: Determining the Strain Tensor in the Skin ($\boldsymbol{\varepsilon}$)

For a stretch in the X axis by $\varepsilon_{xx} = 0.30$, the transverse strains determined from the Poisson effect are:

$$\varepsilon_{yy} = \varepsilon_{zz} = -\nu \cdot \varepsilon_{xx} = -0.48 \cdot 0.30 = -0.144 \quad (-14.4\%)$$

Therefore, the strain tensor in the silicone has the form:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} 0.30 & 0 & 0 \\ 0 & -0.144 & 0 \\ 0 & 0 & -0.144 \end{bmatrix}$$

Step 2: Determining Stresses in the Silicone ($\boldsymbol{\sigma}$)

Using the previously determined Lamé constants for silicone:

$$\mu = \frac{E}{2(1+\nu)} = \frac{2}{2(1.48)} \approx 0.676 \text{ MPa}$$

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} = \frac{2 \cdot 0.48}{(1.48)(0.04)} \approx 16.216 \text{ MPa}$$

Trace of the strain tensor (volume change):

$$\text{tr}(\boldsymbol{\varepsilon}) = 0.30 - 0.144 - 0.144 = 0.012$$

We determine the principal stress along the X axis:

$$\sigma_{xx} = \lambda \text{tr}(\boldsymbol{\varepsilon}) + 2\mu \varepsilon_{xx} = 16.216(0.012) + 2(0.676)(0.30) \approx 0.1946 + 0.4056 = 0.6002 \text{ MPa}$$

Step 3: Safety and Delamination Analysis

1. Will the silicone tear?

The maximum stress $\sigma_{xx} = 0.60 \text{ MPa}$ is significantly smaller than the strength of silicone ($\sigma_{ult} = 8 \text{ MPa}$). **The silicone will easily withstand this stretching.**

2. Will the copper detach (delaminate) from the silicone?

Due to the gigantic difference in stiffness ($E_{Cu}/E_{PDMS} \approx 55000$), a shear stress σ_{xz} arises at the phase boundary. From FEM calculations, we know that at the edge of the track, the shear stress reaches a value of $\tau_{max} \approx 2.5 \text{ MPa}$. If the adhesion of the glue/substrate is lower than 2.5 MPa , the track will peel off the skin, causing device failure.